

MODIM CAPSULE

CONTENT

Each capsule contains Loperamide hydrochloride 2 mg.

DESCRIPTION

White granules in green-light green no.4 capsule, imprinted with "SPS" in white colour on cap and body.

INDICATION

Modim capsule is used in the symptomatic treatment of acute and chronic diarrhoea. In patients with ileostomies, **Modim** capsule may be used to control the frequency and fluidity of the stools.

RECOMMENDED DOSAGE

Adults and Children 6 to 17 years

Acute Diarrhoea: Initial dose of 2 capsules (4 mg) for adults and 1 capsule (2 mg) for children, followed by 1 capsule (2 mg) after each loose stool.

Chronic Diarrhoea: Initial dose of 2 capsules (4 mg) daily for adults and 1 capsule (2 mg) daily for children. Initial dose should be adjusted until 1 – 2 solid stools a day are obtained. This is usually achieved with a maintenance dose of 1 – 6 capsules (2 mg – 12 mg) daily.

Maximum dose for acute and chronic diarrhoea is 8 capsules (16 mg) daily for adults; in children, it must be related to the body weight (3 capsules / 20 kg) but should not exceed a maximum of 8 capsules per day.

Elderly

No dose adjustment is required for the elderly.

Renal impairment

No dose adjustment is required for patients with renal impairment.

ROUTE OF ADMINISTRATION

For oral administration. Capsules should be taken with liquid.

CONTRAINDICATIONS

Modim capsules should not be taken in:

- Patients with known hypersensitivity to loperamide hydrochloride or to any of the excipients.
- Children under 6 years of age.
- When inhibition of peristalsis is to be avoided due to the possible risk of significant sequelae including ileus, megacolon and toxic megacolon.

Modim capsules should not be used as the primary therapy in:

- Patients with acute dysentery, which is characterized by blood in stools and high fever.
- Patients with acute ulcerative colitis.
- Patients with bacterial enterocolitis caused by invasive organisms including *Salmonella*, *Shigella*, and *Campylobacter*.
- Patients with pseudomembranous colitis associated with the use of broad-spectrum antibiotics.

WARNINGS AND PRECAUTIONS

Modim capsule is not recommended for children under 6 years of age. The use of loperamide has been associated with fatal episodes of paralytic ileus in infants and young children.

Appropriate fluid and electrolyte therapy should be given to protect against dehydration in all cases of diarrhoea. Oral rehydration therapy - which is the use of appropriate fluids including oral rehydration salts - remains the most effective treatment for dehydration due to diarrhoea. The intake of as much of these fluids as possible is therefore imperative.

Drug-induced inhibition of peristalsis may result in fluid retention in the intestine, which may aggravate and mask dehydration and depletion of electrolytes. If severe dehydration or electrolyte imbalance is present, administration of **Modim** capsule should be withheld until appropriate corrective therapy has been initiated. Treatment of diarrhoea with loperamide hydrochloride is only symptomatic. Whenever an underlying aetiology can be determined, specific treatment should be given when appropriate. In acute diarrhoea, if clinical improvement is not observed within 48

hours, the administration of loperamide hydrochloride should be discontinued and patients should be advised to consult their doctor. Patients with AIDS treated with loperamide hydrochloride for diarrhoea should have therapy stopped at the earliest signs of abdominal distension. There have been isolated reports of constipation with an increased risk for toxic megacolon in AIDS patients with infectious colitis from both viral and bacterial pathogens treated with loperamide hydrochloride.

Although no pharmacokinetic data are available in patients with hepatic impairment, loperamide hydrochloride should be used with caution in patients with hepatic impairment due to reduced first pass metabolism. This may result in a relative overdose leading to CNS toxicity.

The use of higher than the recommended doses for control of the diarrhoea may cause abnormal heart rhythms and serious cardiac events leading to death. However, in adult patients receiving the recommended dosage of loperamide, cases of syncope and ventricular tachycardia have been reported. Some of these patients were taking other drugs or had other risk factors that may have increased their risk of cardiac adverse reactions.

Abuse and misuse of loperamide, as an opioid substitute, have been described in individuals with opioid addiction (see *Symptoms and Treatment of Overdose*).

Effects on ability to drive and use machines

Patients may experience tiredness, dizziness, or drowsiness may occur in the setting of diarrheal syndromes treated with loperamide hydrochloride. Therefore, it is advisable to use cautiously when driving or operating machinery.

DRUG INTERACTIONS

Quinidine or Ritonavir (P-glycoprotein inhibitors)

Concomitant administration will result in a 2 to 3-fold increase in loperamide plasma levels.

Saquinavir

Concomitant administration decreases saquinavir plasma concentrations, and increased loperamide plasma concentrations.

Itraconazole

Itraconazole is an inhibitor of CYP3A4 and P-glycoprotein. Concomitant administration will result in a 3 to 4-fold increase in loperamide plasma concentrations.

Gemfibrozil

Concomitant administration increases loperamide plasma concentrations by approximately 2-fold.

Itraconazole and Gemfibrozil

Combination of both medications with loperamide increase peak plasma levels of loperamide by 4-fold and 13-fold increase in total plasma exposure.

Ketoconazole

Ketoconazole is an inhibitor of CYP3A4 and P-glycoprotein. Concomitant administration will result in a 5-fold increase in loperamide plasma concentrations.

Oral Desmopressin

Concomitant administration results in a 3-fold increase of desmopressin plasma concentrations, presumably due to slower gastrointestinal motility.

St. John's Wort

Interaction leads to patient experiencing delirium with symptoms of confusion, agitation and disorientation.

Valerian

Interaction leads to patient experiencing delirium with symptoms of confusion, agitation, and disorientation.

Co-trimoxazole

Concomitant administration increases bioavailability of loperamide by inhibiting its first-pass metabolism.

It is expected that drugs with similar pharmacological properties may potentiate loperamide's effect and that drugs that accelerate gastrointestinal transit may decrease its effect.

PREGNANCY AND LACTATION

Pregnancy

Loperamide is poorly absorbed from the gut. Available data indicates minimum risk of malformation when used during pregnancy. However, therapeutic benefits should be weighed against potential risk to the foetus when given during pregnancy, especially during the first trimester.

Lactation

Small amounts of loperamide may appear in human breast milk and therefore should not be recommended during breast-feeding.

SIDE EFFECTS

Frequency	System and Indication
Common	<u>Nervous system disorders</u> - Headache <u>Gastrointestinal disorders</u> - Constipation - Nausea - Flatulence
Uncommon	<u>Nervous system disorders</u> - Dizziness - Somnolence <u>Gastrointestinal disorders</u> - Abdominal pain - Abdominal discomfort - Dry mouth - Upper abdominal pain - Vomiting - Dyspepsia <u>Skin and subcutaneous tissue disorders</u> - Rash
Rare	<u>Immune system disorders</u> - Hypersensitivity reaction - Anaphylactic reaction - Anaphylactoid reaction <u>Nervous system disorders</u> - Loss of consciousness - Stupor - Coordination abnormality - Hypertonia <u>Eye disorders</u> - Miosis <u>Gastrointestinal disorders</u> - Ileus - Megacolon - Abdominal distension <u>Skin and subcutaneous tissue disorders</u> - Bullous eruption (Including Stevens-Johnson syndrome, Toxic epidermal necrolysis and Erythema multiforme) - Angioedema - Urticaria - Pruritus <u>Renal and urinary disorders</u> - Urinary retention <u>General disorders</u> - Fatigue
Post-marketing experience (Frequency 'not known')	<u>Cardiac disorders</u> - QT/QTc interval prolongation, Torsades de Pointes, other ventricular arrhythmias, cardiac arrest, syncope, and death (see <i>Warnings and Precautions</i>). <u>Gastrointestinal disorders</u> - Acute pancreatitis

SYMPTOMS AND TREATMENT OF OVERDOSAGE

Common overdose effects: miosis, drowsiness, nausea, vomiting, abdominal pain and headache. Seizures, dystonic reactions, ileus and pancreatitis have also occurred following overdose.

May produce CNS depression; supportive treatment should be given and if respiratory depression occurs. Naloxone is an effective reversal agent. Initial dose of 2.0 mg intravenously; repeat as necessary.

Abuse, misuse and/or overdose with excessively large doses of loperamide, may unmask Brugada syndrome.

In individuals who have intentionally ingested overdoses (reported in doses from 40 mg up to 792 mg per day) of loperamide hydrochloride, prolongation of the QT/QTc interval, Torsades de Pointes, other ventricular arrhythmias and cardiac arrest, have been observed (see *Warnings and Precautions*). Fatal cases have also been reported.

PHARMACODYNAMICS

Loperamide is a synthetic derivative of pethidine that inhibits gut motility and may also reduce gastrointestinal secretions.

Loperamide binds to the opiate receptor in the gut wall. Consequently, it inhibits the release of acetylcholine and prostaglandins, thereby reducing propulsive peristalsis, and increasing intestinal transit time.

Loperamide increases viscosity, increases bulk density, reduces daily faecal volume, and diminishes loss of fluids and electrolytes. Loperamide increases the tone of the anal sphincter, thereby reducing incontinence and urgency. Additionally, loperamide will prolong mouth-to-cecum transit time without affecting gastric emptying.

PHARMACOKINETICS

Absorption: Most ingested loperamide is absorbed from the gut, but as a result of significant first pass metabolism, systemic bioavailability is only approximately 0.3%.

Distribution: Little intact drug reaches the systemic circulation.

Gastrointestinal tract: 85%

Liver: 5%

Tissues 0.04 to 0.2%

The plasma protein binding of loperamide is 95%, mainly to albumin.

Metabolism: Loperamide undergoes significant first pass metabolism in the liver. Oxidative N-demethylation is the main metabolic pathway for loperamide, and is mediated mainly through CYP3A4 and CYP2C8. Oxidative n-dealkylation may be another pathway. Due to this very high first pass effect, plasma concentrations of unchanged drug remain extremely low.

Elimination: The half-life of loperamide in man is 10.8 hours with a range of 9 – 14 hours. Approximately 50% of an orally administered dose is excreted unchanged, primarily in the faeces; there is slight urinary excretion.

LIST OF EXCIPIENTS

Lactose monohydrate, cornstarch, sodium starch glycolate, talcum, magnesium stearate

PRESENTATION

Box of 50 x 10's capsules

STORAGE CONDITION

Store in a cool dry place, below 30°C.

Keep out of the reach of children.

Jauhi daripada kanak-kanak.

SHELF-LIFE: Please refer to the outer box label.

MANUFACTURED BY:



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